

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 12

DUE: WEDNESDAY, APRIL 29, 2026, BEFORE 23:59PM

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let $A \subset \mathbb{R}^n$ be a compact subspace. Let $U_- = \mathbb{R}^{n+1} \setminus ([0, \infty) \times A)$ and $U_+ = \mathbb{R}^{n+1} \setminus ((-\infty, 0] \times A)$.
 - (a) Briefly explain why U_+ and U_- are both contractible. (*For instance, define a homotopy inverse to the inclusion $\{1\} \times \mathbb{R}^n \hookrightarrow U_+$, and similarly for U_- .*) Then, briefly explain why $U_- \cap U_+$ is homotopy equivalent to $\mathbb{R}^n \setminus A$.
 - (b) Use the Mayer–Vietoris sequence in reduced homology to construct an isomorphism

$$\tilde{H}_{k+1}(\mathbb{R}^{n+1} \setminus (\{0\} \times A)) \rightarrow \tilde{H}_k(\mathbb{R}^n \setminus A)$$

for all k .

2. *This question is a continuation of the previous.*
 - (a) Deduce an isomorphism $\tilde{H}_k(\mathbb{R}^n \setminus A) \rightarrow \tilde{H}_{k+m}(\mathbb{R}^{n+m} \setminus (\{0\} \times A))$ for any k and any m .
 - (b) Let $B \subset \mathbb{R}^m$ be another compact subset, and assume that there exists a homeomorphism $\phi : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^{n+m}$ such that $\phi(\{0\} \times A) = B \times \{0\}$. Deduce isomorphisms

$$\tilde{H}_{n-k}(\mathbb{R}^n \setminus A) \xrightarrow{\cong} \tilde{H}_{m-k}(\mathbb{R}^m \setminus B)$$

for all k .

Remark, not part of the question: it can be proved that such a homeomorphism ϕ always exists, provided A and B are homeomorphic. Using this, it follows that if $j : C \rightarrow \mathbb{R}^n$ is an embedding of a compact topological space C , then the reduced homology of $\mathbb{R}^n \setminus j(C)$ is independent of the embedding, at least up to isomorphism.

3. Let X be a topological space, $U_s \subset X$ open subsets for each $s \in \mathbb{N}$, and assume that $U_s \subset U_{s+1}$ for all $s \in \mathbb{N}$, and that $\bigcup_s U_s = X$.

As sketched in class, it follows from compactness of Δ^p that: for any $c \in C_p(X)$, there exists an $s \in \mathbb{N}$ such that $c \in C_p(U_s) \subset C_p(X)$. This fact may be used without proof in the following.

- (a) For each $s \in \mathbb{N}$, let $i_s : U_s \hookrightarrow X$ denote the inclusion map. Let $x \in H_p(X)$. Prove that there exists an $s \in \mathbb{N}$ and a $y \in H_p(U_s)$ such that $(i_s)_*(y) = x$.
 - (b) For $s \leq t$, let $j_{s,t} : U_s \rightarrow U_t$ denote the inclusion map. Prove that if $y, y' \in H_p(U_s)$ satisfied $(i_s)_*(y) = (i_s)_*(y')$, then there exists a $t \in \mathbb{N}$ such that $(j_{s,t})_*(y) = (j_{s,t})_*(y')$.
4. Let (X, A) be a good pair, and write X/A for the quotient space (i.e., collapse A to a one-point space A/A). Assume that $H_n(X)$ and $H_n(A)$ are finitely generated abelian groups for all $n \in \mathbb{N}$, and that there exists an $N \in \mathbb{N}$ such that $H_n(X) \cong 0$ and $H_n(A) \cong 0$ for all $n > N$.
- (a) Prove that $H_n(X, A)$ is a finitely generated abelian group for all $n \in \mathbb{N}$ and that there exists an $N' \in \mathbb{N}$ such that $H_n(X/A) \cong 0$ for all $n > N'$.
 - (b) Find a formula for $\chi(X/A)$ in terms of $\chi(X)$ and $\chi(A)$.

5. *This question is a continuation of the previous*

Let X be a topological space for which $\chi(X)$ is defined. Let SX denote the suspension of X , as in HW9.

- (a) Prove that $\chi(SX)$ is defined, and $\chi(SX) = 2 - \chi(X)$.
- (b) Let $z = (1, 0) \in S^1$. It may be used without proof that $(S^1 \times X, \{z\} \times X)$ is a good pair, and that $(S^1 \times X)/(\{z\} \times X)$ is homeomorphic to $SX/S\emptyset$. Prove that $\chi(S^1 \times X) = 0$.